

VIKRAMADITYA SEN

Principal Software Architect | Distributed Systems & Cloud Infrastructure

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PROFESSIONAL SUMMARY

Accomplished Principal Software Architect and Engineering Leader with 12+ years of experience designing high-throughput distributed architectures, resilient fault-tolerant systems, and enterprise cloud migrations. Proven track record leading multi-disciplinary engineering organizations, driving microservices transitions handling 45K+ RPS, and architecting mission-critical financial platforms managing billions in transactional throughput with 99.999% availability.

TECHNICAL SKILLS

Architecture & Systems: Distributed Systems, Microservices, Event-Driven, Raft Consensus, High Availability, Fault Tolerance

Cloud & Infrastructure: Kubernetes, AWS, GCP, Docker, Terraform, CI/CD, eBPF, Prometheus, Grafana

Languages: Go, Rust, Java, Scala, Python, C++, SQL, gRPC/Protobuf

Data & Messaging: Apache Kafka, Apache Flink, PostgreSQL, DynamoDB, Redis, Cassandra

WORK EXPERIENCE

Principal Software Architect — Stripe Inc.

San Francisco, CA | 2021 – Present

- Architected next-generation payment routing engine handling 45,000 requests/sec with a 99.999% SLA and zero single points of failure.
- Led company-wide migration from monolithic Ruby on Rails to Golang/gRPC microservices, slashing p99 latency by 64%.
- Formulated real-time streaming anomaly detection pipelines processing \$12B+ in annual transaction volume via Kafka and Apache Flink.
- Mentored 14 Staff/Senior engineers and instituted cross-organization architectural RFC standards across 8 engineering squads.

Senior Engineering Lead — Uber Technologies

Seattle, WA | 2017 – 2021

- Spearheaded global driver dispatch microservice platform operating across 600+ cities in 65 countries with sub-100ms response targets.
- Designed geospatial indexing algorithms leveraging H3 hierarchical spatial indexing, boosting matching efficiency and accuracy by 22%.
- Governed \$4.5M annual AWS cloud budget, generating 31% cost reduction via Kubernetes auto-scaling and spot instance optimization.
- Executed zero-downtime horizontal database partitioning strategy across 20TB+ production PostgreSQL relational clusters.

Senior Software Engineer — Twitter Inc.

Seattle, WA | 2014 – 2017

- Engineered distributed event-bus pipelines processing 500M+ daily tweets using Apache Kafka, Finagle, and Scala.
- Re-architected notification delivery topology, eliminating processing bottlenecks and reducing queue backlog latency by 75%.
- Standardized enterprise OAuth2 and SAML authentication protocols across tier-1 internal engineering infrastructure.

KEY PROJECTS

Distributed Raft Consensus Key-Value Store | Rust, Tokio, gRPC, Protobuf, Jepsen

- Implemented production-ready Raft consensus engine with log compaction, snapshotting, and leader election validated by Jepsen fault-injection tests.

eBPF-Based OpenTelemetry Performance Tracer | Go, eBPF, OpenTelemetry, Prometheus

- Built low-overhead kernel socket metric tracing daemon without SDK overhead, adopted by 2,000+ open-source developers (1.4k GitHub stars).

EDUCATION

M.S. in Computer Science

Stanford University | 2010 – 2012

B.S. in Computer Engineering

University of Washington | 2006 – 2010

PATENTS & PUBLICATIONS

US Patent 10,482,192: Dynamic geospatial routing & dispatch partitioning in distributed environments.

IEEE TPDS: High-throughput fault-tolerant consensus in geographically distributed datacenters.